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Phytoplankton biomass dynamics in the Arabian Sea from VIIRS observations

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ABSTRACT

Using chlorophyll-a (Chl-a) as an indicator for phytoplankton biomass, observations from the Visible Infrared Imaging Radiometer Suite (VIIRS) onboard the Suomi National Polar-orbiting Partnership (SNPP) between 2012 and 2019 in combination with wind, various ocean hydrographic data, and nutrient data are used to conduct a comprehensive study to characterize and quantify the phytoplankton biomass dynamics in the Arabian Sea, and assess the mechanisms that drive the summer and winter phytoplankton blooms. Phytoplankton biomass dynamics in the Arabian Sea are driven by the seasonal reversal of the monsoon wind. The summer phytoplankton bloom located mainly in the western and central Arabian Sea is stronger than the winter phytoplankton bloom in the northern Arabian Sea in terms of both Chl-a values and the extent of phytoplankton bloom coverage. Interannual variability of Chl-a is less significant in comparison to the seasonal variability. Chl-a data measured by the Sea-viewing Wide Field-of-View Sensor (SeaWiFS) from 1997 to 2010 are also used. The comparison of Ch-a climatology data between SeaWiFS and VIIRS shows no long-term trend of Chl-a change in the Arabian Sea. This study provides an improved understanding (with new knowledge) of (1) phytoplankton biomass dynamics, (2) physical, biological, and biogeochemical processes, and (3) nutrient dynamics in the Arabian Sea. The hydrographic and nutrient data reveal two different driving mechanisms for phytoplankton blooms in the summer and winter monsoons. During the summer monsoon, the cold high-nutrient waters are brought to the surface from the thermocline depth through strong coastal upwelling and Ekman pumping, leading to increased Chl-a at the surface-layer. The vertical velocity reaches ~1 m/day at 70 m depth. In comparison, the winter phytoplankton bloom is driven by the moderate boost of the nutrient level due to the mixed-layer deepening and strong vertical mixing caused by the surface cooling in the northern Arabian Sea. The nutrient dynamics in the Arabian Sea also suggests that the nitrate concentration is the major nutrient driver for its seasonal phytoplankton biomass dynamics.

1. Introduction

The Arabian Sea (Fig. 1) is featured with a semiannual reversal of the monsoon. The reversal of the monsoon is caused by the land-sea latent heat difference. In the western Indian Ocean, the Southwest (SW) monsoon from June to September is much stronger than the Northeast (NE) monsoon in the winter (Bruce, 1983). Along the Somali coast and the Arabian Peninsula coast, a low-level jet extends across the Arabian Sea (Findlater, 1969). The wind speed can reach over 15 m/s during the SW monsoon. In the inter-monsoon period, the wind is generally weak.

The dynamics and thermodynamics in the Arabian Sea are driven by the seasonal reversal of the SW and NE monsoons. The monsoon current is also seasonally reversing. During the SW monsoon, the strong alongshore Somalia Current flows from SW to NE in the western Arabian Sea driven by the local forcing and remote forcing (Beal et al., 2013; Schott and McCreary, 2001; Shankar et al., 2002). In the eastern Arabian Sea, the surface current flows out of the sea. The net transport in the upper 500 m can reach about $10^7 m^3/s$ southward out of the Arabian Sea (Shi et al., 1999). During the winter season, the flow is weak and surface water circulation reverses the direction. The net transport in the upper 500 m is northward at about 5×10^6 m^3/s (Shi et al., 1999). In the Arabian Sea, the sea surface salinity (SSS) trends lower from north to

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south. During the winter, the excess of evaporation over precipitation drives the convective mixing in the northern Arabian Sea, and helps form the Arabian Sea High-Salinity Water (ASHSW), which spreads southward along a 24 sigma-t surface (Kumar and Prasad, 1999).

During the SW monsoon, the strong alongshore monsoon winds lead to strong costal upwelling along the Somali coast and the Arabian Peninsula coast. Significant sea surface temperature (SST) drop can be observed along the coastal regions during the SW monsoon owing to entrainment of cool subsurface water at the coast and its subsequent offshore advection (Mccreary and Kundu, 1989; Rao and Goswami, 1988). The upwelling front gradually moves 100 km offshore with the development of the SW monsoon (Shi et al., 2000). In the region of 1000 km alongshore and 400 km offshore, 8×10^6 m^3 s^{-1} were upwelling from a depth of 50 m to the surface (Smith and Bottero, 1977). From the satellite altimetry measurements, it was estimated that the upwelling velocity at the 100 m depth was $\sim 2 \times 10^{-5}$ m/ s ($\sim$1.7 m/day) in the region with 260 km offshore distance along the Arabian Peninsula coast during the SW monsoon in 1993, 1994, and 1995 (Shi et al., 2000). The upwelled water was also found to advect in the form of plumes (Manghnani et al., 1998). During the NE monsoon, the decrease of the temperature in the mixed-layer is driven by the weakening solar radiation, stronger evaporative cooling, as well as the entrainment mixing in the northern Arabian Sea (Mccreary and Kundu, 1989). The deepest mixed-layer depth was also observed during the NE monsoon (Morrison et al., 1998).

The nutrient fields in the Arabian Sea are highly related to the seasonal monsoon and various ocean processes. The euphotic zone of the northern Arabian Sea is generally $\sim$40 m, ranging between 20 and 60 m. During the SW monsoon, deep, cold, and high nutrient water with temperature of 24.2 °C was observed to be brought up to the surface. In the inshore regions, the NO_3 was elevated to 18 μmol l^{-1} and PO_4 reached 1.48 μmoll^{-1} (Woodward et al., 1999). Inorganic nitrogen-to-phosphate ratios in the Arabian Sea suggests that inorganic nitrogen is a more limiting nutrient for phytoplankton growth than the inorganic phosphate (Morrison et al., 1998). Widely spread moderate nutrient water can be found in the surface layer in the Arabian Sea during the NE monsoon (Morrison et al., 1998). In the Arabian Sea, the oxygen minimum zone (OMZ) is located at the depth 150–1200 m with the oxygen less than 4.5 μmol l^{-1} (Morrison et al., 1999). The OMZ intensified and deepened in the Arabian Sea in the last several decades with the increase of the monsoon wind intensity (Lachkar et al., 2018; Naqvi et al., 2009). The denitrification process in the OMZ is the dominant nitrogen process in the Arabian Sea (Ward et al., 2009). In fact, the loss of nitrogen is balanced by the advective input of nitrogen from the south (Bange et al., 2000).

The Arabian Sea is one of the most productive regions in the world ocean (Ryther and Menzel, 1965). It accounts for nearly 30% of the world ocean nitrogen loss (Bange et al., 2005; Lam et al., 2011; Naqvi et al., 2006; Paulmier and Ruiz-Pino, 2009; Ward et al., 2009). Furthermore, the Arabian Sea is becoming more productive with the warming of the Eurasian landmass (Goes et al., 2005). In the Arabian Sea, there are two peaks of phytoplankton chlorophyll-a (Chl-a). One occurs in the SW monsoon (summer) and another in the NE monsoon (winter). Both Chl-a peaks are related to the mixed-layer deepening and concomitant-enhanced nutrient supplies (Banse, 1987). The Chl-a concentration in the Arabian Sea can also be influenced by the tropical cyclones (Subrahmanyam et al., 2002). In fact, Chl-a and its variability not only represent biological processes, but also are related to the nutrient dynamics and modulate ocean physical processes such as SST (Nakamoto et al., 2000) and Rossby wave (Amol et al., 2020).

The phytoplankton structure is linked to the hydrographic characteristics, the environmental driving forcing, and ocean mesoscale features (Shalapyonok et al., 2001). During the NE monsoon, the injection of the nutrient up into the surface layer due to the convection process, which is driven by the surface cooling, leads to a high productivity in the northern Arabian Sea (Madhupratap et al., 1996). In addition to the phytoplankton blooms, other biological properties, such as the colored dissolved organic matter (CDOM), are highly related to the SST, and the production of CDOM is associated with the phytoplankton bloom in the SW monsoon (Coble et al., 1998). The different spatial distributions of

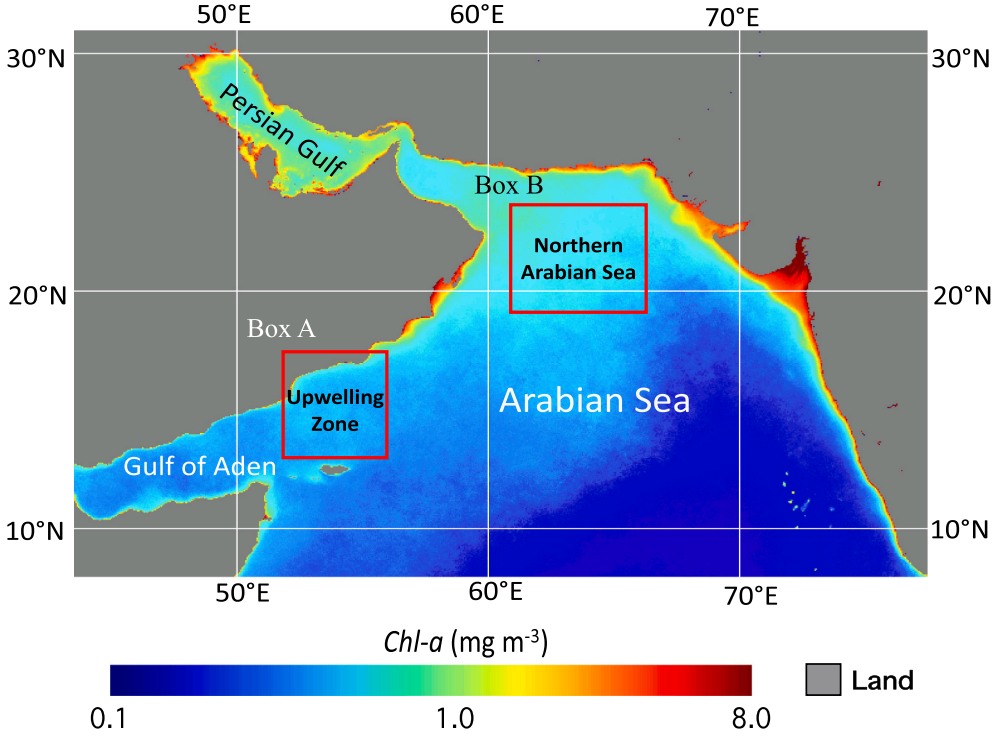


Fig. 1. (a) Climatology Chl-a in the Arabian Sea (43.5°E–77.5°E, 8°N–31°N.) between 2012 and 2019. Note that Box A (52.13°E–56.03°E, 13.03°N–17.28°N) for the summer SW monsoon and Box B (61.68°E–66.83°E, 18.88°N–23.48°N) for the winter NE monsoon are marked in Fig. 1 for further analysis.

CDOM during the SW and NE monsoons reflect the different wind regimes in these two monsoons (Del Castillo and Coble, 2000). During the inter-monsoon periods, the Arabian Sea relaxes with weak wind fields, and the region transitions to an oligotrophic condition with Chl-a much lower than those during the two monsoons. This shows a significant contrast to the biogeochemical processes during the SW and NE monsoons (Wiggert et al., 2005).

In this study, a variety of in situ and satellite data, including surface winds, satellite ocean color data, SST, monthly model outputs of the potential temperature, salinity, horizontal and vertical velocity, as well as the monthly climatological nutrient profile data in the Arabian Sea, are used to conduct a comprehensive study to characterize and quantify the phytoplankton blooms in the SW monsoon in summer and NE monsoon in winter. The seasonal and interannual variability of phytoplankton blooms in the region and its optical features are assessed using satellite measurements (2012–2019) from the Visible Infrared Imaging Radiometer Suite (VIIRS) onboard the Suomi National Polar-orbiting Partnership (SNPP). In addition, Chl-a data measured by the Sea-viewing Wide Field-of-view Sensor (SeaWiFS) in the region from 1997 to 2010 are also used. SeaWiFS Chl-a data provided useful water optical and biological properties in the Arabian Sea in the early years. The ocean physical processes, which are represented as SST, salinity profiles, velocity fields, and mixed-layer depth (MLD), are explored to address the linkage between the two phytoplankton blooms and the physical processes driven by the seasonal monsoons. In addition, the nutrient dynamics in the Arabian Sea are further examined to show how the physical processes in the region impact the nutrient supplies at the surface layer and eventually drive the phytoplankton blooms in the summer and winter seasons.

2. Data and methods

2.1. VIIRS- and SeaWiFS-derived Chl-a products

The VIIRS-SNPP was launched in late 2011 to provide operational satellite products for global atmosphere, land, and ocean following the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard the Terra and Aqua satellites (Goldberg et al., 2013). For the VIIRS ocean color products, the normalized water-leaving radiance spectra $nL_w(\lambda)$ at the visible wavelengths of 410, 443, 486, 551, 671 nm and at the near-infrared (NIR) wavelengths of 745 and 862 nm with spatial-resolution of 750 m (Wang et al., 2013), as well as $nL_w(\lambda)$ at 638 nm with spatial-resolution of 375 m (Wang and Jiang, 2018), are derived using the NIR and shortwave infrared (SWIR) combined atmospheric correction algorithms (Gordon and Wang, 1994; Wang, 2007; Wang and Shi, 2007; Wang et al., 2009b). The on-orbit vicarious calibration (Wang et al., 2016) was carried out using the in situ $nL_w(\lambda)$ measurements from the Marine Optical Buoy (MOBY) (Clark et al., 1997) to produce high quality global satellite ocean color products.

VIIRS-measured $nL_w(\lambda)$ spectra are the fundamental products in satellite ocean color remote sensing (IOCCG, 2010). Indeed, all other optical, biological, and biogeochemical products such as Chl-a concentration, diffuse attenuation coefficient at 490 nm $K_d(490)$ (Wang et al., 2009a), suspended particulate matter (SPM) (Wei et al., 2021; Yu et al., 2019), etc., are derived from satellite-measured $nL_w(\lambda)$ spectra. As the primary ocean color product for VIIRS satellite observations, Chl-a data are produced using the blended Chl-a algorithms for the global ocean (Hu et al., 2012; O'Reilly et al., 1998; O'Reilly and Werdell, 2019; Wang and Son, 2016). For oligotrophic waters, Chl-a are computed with the color index (CI)-based algorithm (Hu et al., 2012; Wang and Son, 2016). The CI-based Chl-a algorithm is further merged with the blue-green reflectance ratio algorithm (O'Reilly et al., 1998; O'Reilly and Werdell, 2019) to produce Chl-a data for the global open ocean and coastal/inland waters (Hu et al., 2012; Wang and Son, 2016).

SeaWiFS was launched in 1997 and was active between September 1997 and December 2010. The spectral band specification in the visible and NIR is similar to that of VIIRS. Global satellite Chl-a products have been routinely produced from various missions in the US, e.g., SeaWiFS, MODIS, and VIIRS. The Chl-a algorithm for SeaWiFS was mainly based on the Chl-a algorithm developed in 1998 (O'Reilly et al., 1998). This algorithm was further tuned, improved, and adjusted for the other satellite ocean color sensors (Hu et al., 2012; O'Reilly and Werdell, 2019; Wang et al., 2020; Wang and Son, 2016). On the other hand, all $nL_w(\lambda)$ retrievals from these sensors were vicariously calibrated using the MOBY in situ optics measurements (Clark et al., 2002; Franz et al., 2007; Wang et al., 2016). For the global open ocean, the atmospheric correction algorithm is similar for these satellite sensors, based on the Gordon and Wang (1994) algorithm.

In this study, we use the VIIRS-derived Chl-a products (as a surrogate for the phytoplankton biomass) and $nL_w(\lambda)$ spectra to investigate the temporal variations of phytoplankton biomass in the Arabian Sea. For the monthly composites of Chl-a and $nL_w(\lambda)$ spectra, their values are computed as the median values from all valid Chl-a and $nL_w(\lambda)$ retrievals in a certain month between 2012 and 2019. A similar approach is used to calculate Chl-a and $nL_w(\lambda)$ seasonal composites during the SW and NE monsoons.

2.2. Wind data

The reanalyzed surface wind data from the National Center for Environmental Prediction (NCEP)/National Center for Atmospheric Research (NCAR) have a variety of temporal resolutions, including 4-time/day, daily, and monthly, with zonal and meridional winds at the spatial resolution of 2.5°. The past wind data from 1946 to present were assimilated in the state-of-the-art analysis/forecast system known as the NCEP/NCAR Reanalysis (Kalnay et al., 1996). In the wind data, there are 17 levels between the 1000 mb and 10 mb heights. In this study, the monthly wind data at 1000 mb level are used to characterize the wind fields during the SW and NE monsoons, as well as inter-monsoons.

2.3. SST, SSS, vertical velocity (DZ/DT), MLD, salinity, and potential temperature data

To better understand the physical driving forces for the phytoplankton blooms in the Arabian Sea, the output data of the NCEP Global Ocean Data Assimilation System (GODAS) (Behringer et al., 1998; Derber and Rosati, 1989) between 2012 and 2019, corresponding to the VIIRS observations in this study, are analyzed. The GODAS is an ocean analysis and reanalysis system with the spatial resolution of 1/3 × 1/3 degrees and 40 levels. The GODAS is forced by the momentum flux, heat flux, and the fresh water flux from NCEP/NCAR Reanalysis. Both the temperature and salinity profiles are assimilated in the system with a three-dimensional variational data assimilation (3DVAR) scheme (Derber and Rosati, 1989). It was reported that the model results were matched well with in situ ocean observations (Behringer and Xue, 2004; Ravichandran et al., 2013).

In this study, the ocean salinity and potential temperature profiles at water depths of 5, 15, 25, 35, 45, 55, 65, 75, 85, 95, 105, 115, 125, 135, 145, 155, 165, 175, 185, 195, 205, 215, 225, 238, 262, 303, 366, 459 m in the upper 500 m are used. In addition, the vertical velocity at water depths of 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 231, 250, 282, 334, 412, and 521 m are also used to characterize upwelling in the Arabian Sea. The MLD in the World Ocean Atlas 2018 (WOA18) includes approximately 3 million new oceanographic data in the Arabian Sea. The MLD in this dataset is defined as the depth of the buoyancy difference with respect to the surface level < 0.03 (cm s^{-2}).

2.4. Nutrient data

WOA18 is the world's largest publicly available collection of subsurface ocean profile data. The ocean parameters include temperature,

salinity, dissolved oxygen, silicate, nitrate, phosphate, etc. These parameters are objectively analyzed into the domain of 1° grid horizontally and 102 levels vertically at water depth between sea surface and 5500 m. The nitrate and phosphate data are distributed as the global monthly climatology and seasonal climatology. In this study, the monthly climatology nitrate and phosphate data are selected to study the variability of the nitrate and phosphate in the Arabian Sea and their linkage to the phytoplankton blooms during the SW and NE monsoons.

3. Results

3.1. Chl-a in the Arabian Sea

In the Arabian Sea, the climatology Chl-a from VIIRS observations from 2012 to 2019 generally trend lower from north to south (Fig. 1). In the northern end of the Arabian Sea, including the Persian Gulf, Chl-a values are ~1.0 mg m^{-3}. On the western and eastern coast of the Arabian Sea, enhanced Chl-a can be found. In the southeastern Arabian Sea, Chl-a drop to < ~0.2 mg m^{-3}.

Fig. 2 shows the seasonal reversal of the monsoon winds in the Arabian Sea. In January, the region is dominated with the NE winds. The wind speed reaches up to ~10 m/s in the southwestern Arabian Sea. In the spring intermonsoon, winds are weak all over the Arabian Sea (Fig. 2b). The strongest winds can be found during the summer season. The SW monsoon wind reaches over ~15 m/s along the Somalia coast and Arabian Peninsula coast. The winds in the central and eastern Arabian Sea are also strong (Fig. 2c). In the fall intermonsoon (FIM), the wind fields relax significantly and become patternless (Fig. 2d).

Following the seasonal monsoon, Chl-a concentrations in the Arabian Sea show significant seasonal changes (Fig. 3). During the NE monsoon, the phytoplankton bloom occurs in the northern Arabian Sea. In fact, Chl-a in the Arabian Sea reach the maximum of the whole year in March in the northern Arabian Sea (Fig. 3c). Indeed, Chl-a reach over ~2 mg m^{-3} in a vast region of the northern Arabian Sea. After the northern Arabian Sea phytoplankton bloom in March, Chl-a concentrations in the entire Arabian Sea drop remarkably in the spring intermonsoon (Fig. 3d and e). They reach the lowest one of ~0.1 mg m^{-3} in most of the Arabian Sea (Fig. 3e).

With the development of the SW monsoon, Chl-a concentrations in the Arabian Sea show a broad increase (Fig. 3g, h, and i). This is especially true in the western Arabian Sea. Even though there are no Chl-a retrievals due to constant cloud coverage in July and August, the strong phytoplankton bloom in the offshore region of the Arabian Peninsula is remarkable as shown in Fig. 3h and i. The phytoplankton bloom becomes less significant in the western Arabian Sea when the monsoon winds reduced in October, and Chl-a concentrations in the Arabian Sea show the similar spatial distributions in the fall intermonsoon in October (Fig. 3j), November (Fig. 3k), and the start of the NE monsoon in December (Fig. 3l).

Chl-a are the biological products derived from VIIRS-measured $nL_w(\lambda)$ spectra as a function of $nL_w(\lambda)$ ratios in $nL_w(551)/nL_w(443)$ and $nL_w(551)/nL_w(486)$. Therefore, $nL_w(\lambda)$ spectra can provide additional information that can help to better understand the Chl-a variability in the Arabian Sea. Fig. 4 shows the characterization of the climatology $nL_w(\lambda)$ and seasonal changes of $nL_w(\lambda)$ at the wavelengths of 410, 443, and 551 nm. Indeed, the depressed climatology $nL_w(410)$ and $nL_w(443)$ can be observed in the northern Arabian Sea (Fig. 4a and f). The seasonal change of $nL_w(551)$ is not as significant as $nL_w(410)$ and $nL_w(443)$, even though $nL_w(551)$ are slightly higher during the SW monsoon (Fig. 4n) and fall intermonsoon (Fig. 4o). Enhanced $nL_w(410)$ and $nL_w(443)$ are observed in the southern Arabian Sea in April (Fig. 4c and h). This represents more oligotrophic waters in the region, and is consistent with low Chl-a during the spring intermonsoon as shown in Fig. 3e. Even though a large area of the Arabian Sea is constantly covered with clouds, low $nL_w(410)$ and $nL_w(443)$ (Fig. 4d and i) with moderately enhanced $nL_w(551)$ (Fig. 4n) in the western Arabian Sea suggest that the western and central Arabian Sea is dominated by the productive waters and Chl-a concentrations in the region are the highest in summer.

In comparison with the significant seasonal variability of Chl-a and $nL_w(\lambda)$ in the Arabian Sea, the interannual variability of the SW and NE

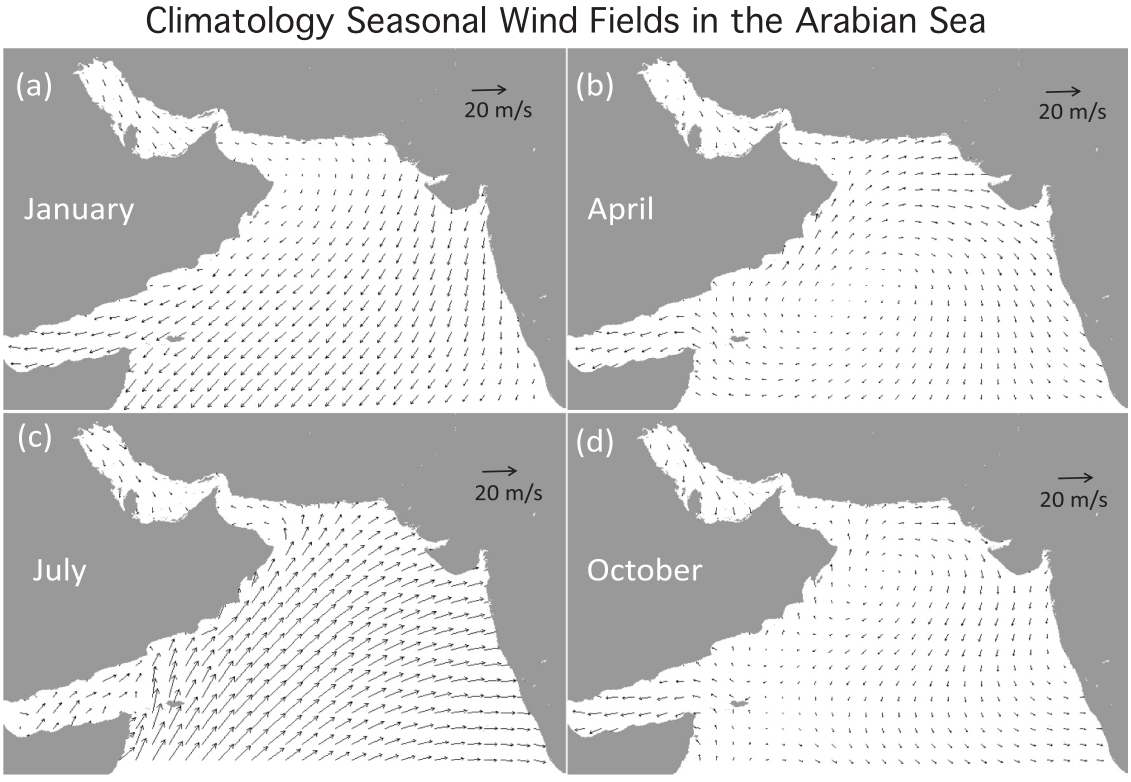


Fig. 2. Climatology seasonal wind fields between 2012 and 2019 in the Arabian Sea for the month of (a) January, (b) April, (b) July, and (d) October.

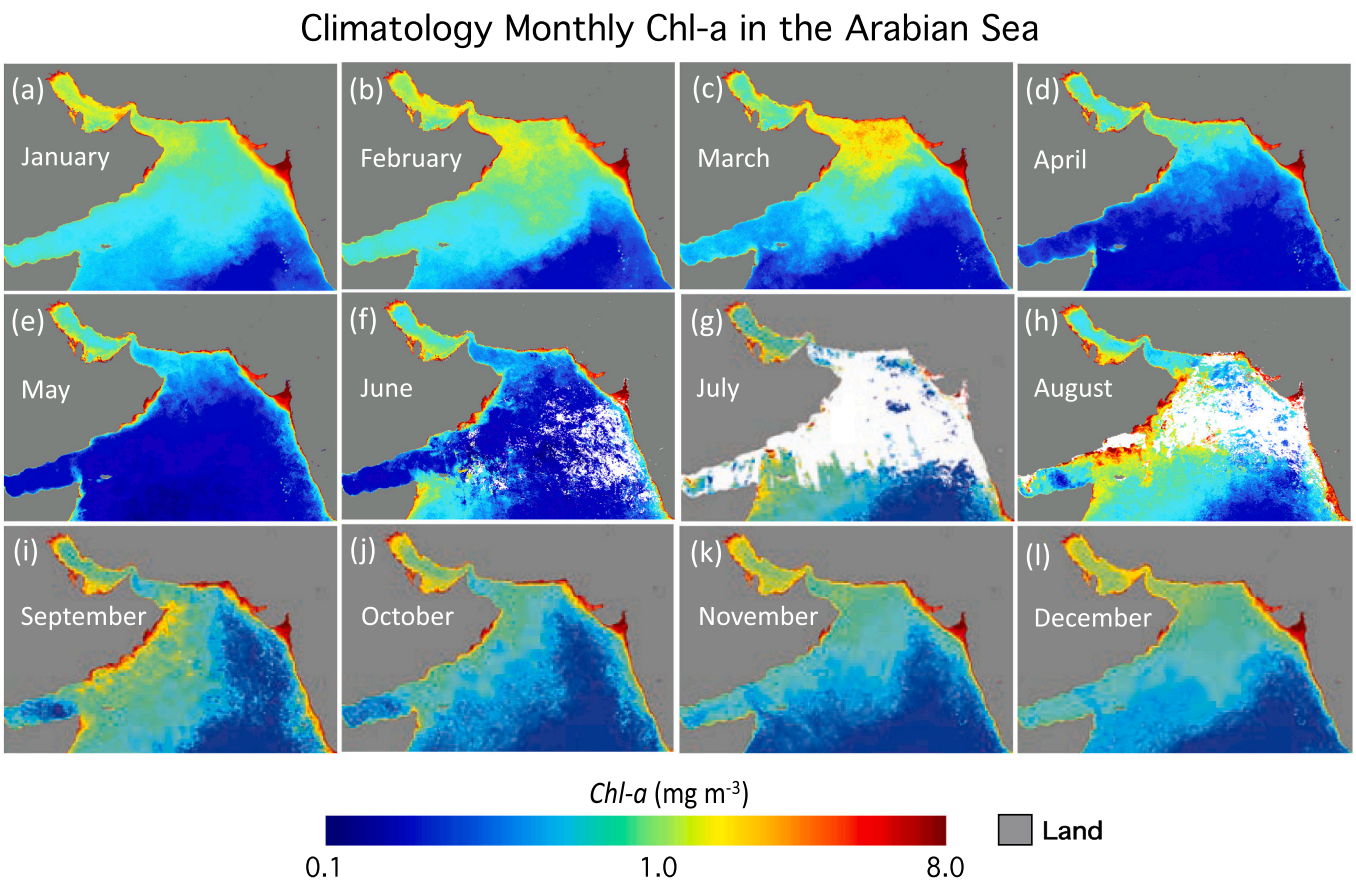


Fig. 3. Climatology monthly Chl-*a* between 2012 and 2019 in the Arabian Sea in (a) January, (b) February, (c) March, (d) April, (e) May, (f) June, (g) July, (h) August, (i) September, (j) October, (k) November, and (l) December.

monsoon phytoplankton blooms are not significant with interannual variability less than ~15% of the seasonal variability for most of the Arabian Sea. Fig. 5 shows annual Chl-a distributions from 2012 to 2019 during the NE monsoon in January–March. Chl-a distributions during the NE monsoon from 2012 to 2019 were similar in terms of the spatial distributions and the magnitude. During the NE monsoon, the region featured elevated Chl-a in the northern Arabian Sea. In fact, Chl-a values reached ~2 mg m^{-3}. On the other hand, low Chl-a ~ 0.1–0.2 mg m^{-3} were located in the southern Arabian Sea. During the NE monsoons in 2012, 2015, and 2017, Chl-a in the northern Arabian Sea were slightly higher than those in the other years. However, these were still insignificant in comparison to the seasonal variability of Ch-a in this region. It is also noted that the moderately depressed Chl-a in the northeastern Arabian Sea during the NE monsoon (Fig. 5e) in 2016 were consistent with the report of the decreasing Chl-a due to the weakening convection associated with the El Niño–Southern Oscillation (ENSO) in 2015–2016 (Vidya and Kurian, 2018).

During the SW monsoon in July–September, the phytoplankton bloom was located in the western and central Arabian Sea (Fig. 6). The Chl-a spatial distribution during the SW monsoon (Fig. 6) was apparently different from that during the NE monsoon (Fig. 5). Significantly enhanced Chl-a concentrations were observed along the Arabian Peninsula coast. During the SW monsoons in 2013 and 2018, more areas in the Arabian Sea were covered with the enhanced Chl-a than that in the other years. During the SW monsoon in 2015, the enhanced Chl-a had the least extent coverage in the Arabian Sea. In 2018, high Chl-a over ~2 mg m^{-3} were observed to extend further to most of the Arabian Sea (Fig. 6g).

3.2. Physical processes in the Arabian Sea

Since the two phytoplankton blooms during the SW and NE monsoons are linked to the monsoon-driven physical processes in the Arabian Sea, it is necessary to study and assess the physical processes and dynamics in the Arabian Sea. SST in the Arabian Sea experienced notable seasonal variation. During the NE monsoon, the current in the region is generally weak (Fig. 7a). SST in the northern Arabian Sea is the lowest in a year due to the surface cooling and the convective mixing in the upper layer, as SST drops to ~24°C in the northern Arabian Sea. This is especially true for waters in the Persian Gulf with SST ~20°C. In April during the spring intermonsoon, SST values in the Arabian Sea show broad increase following the relaxing NE monsoon and reduced sea surface cooling (Fig. 7b). In this month, the current is still weak and featureless in the region.

With the onset of the SW monsoon, strong coastal upwelling and the advection of the coastal upwelling waters result in cool SST in a large area of the western Arabian Sea (Fig. 7c). In the Somali and Arabian Peninsula coastal regions, SST values drop up to ~5°C in comparison to those in April. The circulation in the Arabian Sea is clockwise, and the strong alongshore current with speed over ~0.5 m/s can be found in the western Arabian Sea. The current field in this month also shows that the surface current moves further offshore in the western Arabian Sea and advects the coastal upwelled waters to the central Arabian Sea. During the fall intermonsoon, SST values in the Arabian Sea show a broad increase with the reduced wind fields in the region (Fig. 7d). The current in this period is weak for most part of the Arabian Sea even though the strong current still exists in the offshore region of the Somali coast.

The seasonal variation for the hydrographic parameters of SSS, DZ/DT at the 70 m depth, and MLD is shown in Fig. 8. High SSS dominated

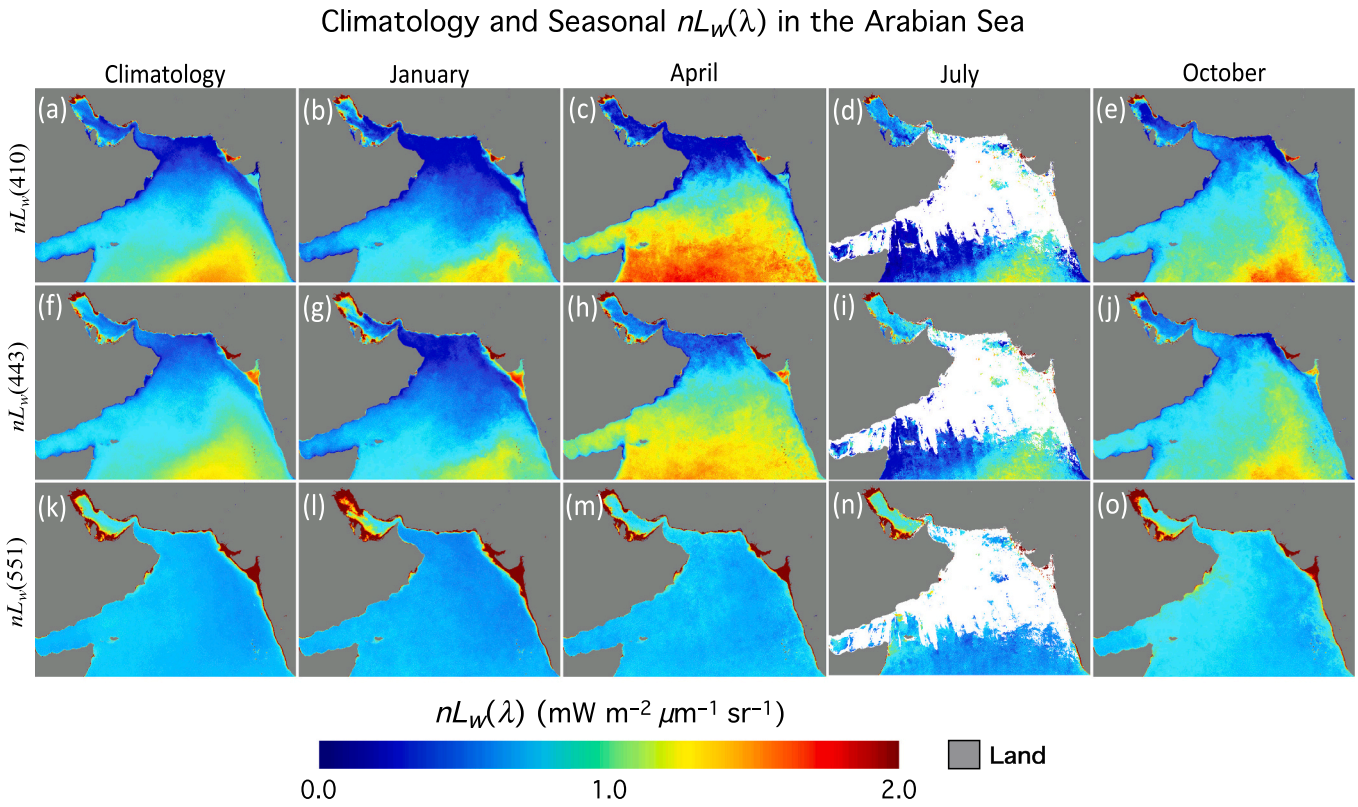


Fig. 4. Climatology and seasonal $nL_w(\lambda)$ in January, April, July, and October in the Arabian Sea for (a–e) $nL_w(410)$, (f–j) $nL_w(443)$, and (k–o) $nL_w(551)$.

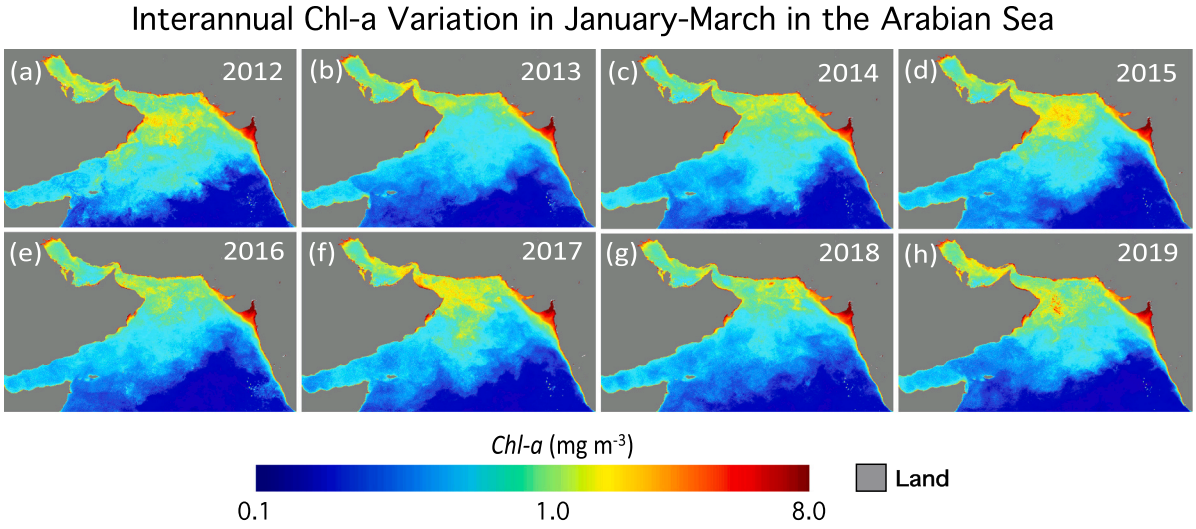


Fig. 5. Chl-a maps during the NE monsoon in January–March in the Arabian Sea for the year of (a) 2012, (b) 2013, (c) 2014, (d) 2015, (e) 2016, (f) 2017, (g) 2018, and (h) 2019.

the northern Indian Sea (Fig. 8a–d) with SSS > 36.5 psu. This could be attributed to the high evaporation and the winter-season convective mixing in the region. In the southern Arabian Sea, SSS are ~35–35.5 psu. However, the seasonal change of SSS is not notable. The vertical velocity at the 70 m depth shows that the vertical movement is small in January (Fig. 8e), April (Fig. 8f), and October (Fig. 8h). On the other hand, strong upwelling can be found along the coastal region of the western Arabian Sea and the offshore region to the left of the Findlater Jet (Findlater, 1969). The vertical velocity at the 70 m depth reaches ~5 m/day. Furthermore, the downwelling velocity to the right of the Findlater Jet can also be ~3–4 m/day and extend from the Somali coast to the central Arabian Sea.

The seasonal change of the MLD in the Arabian Sea is remarkable. In January, the MLD is over 70 m in a large portion of the northern and central Arabian Sea (Fig. 8i). This is attributed to the strong convective mixing in the upper layer due to the surface cooling in the region. In the spring intermonsoon, shallow MLD (Fig. 8j) reflects the strong stratifications with the warming of the surface layer in the Arabian Sea. In this season, the MLD is only about half of that in January. The MLD in the SW monsoon (Fig. 8k) is consistent with the vertical velocity in the Arabian Sea in this season (Fig. 8g). The strong upwelling along the western Arabian Sea coast and the offshore region to the west of the Findlater Jet

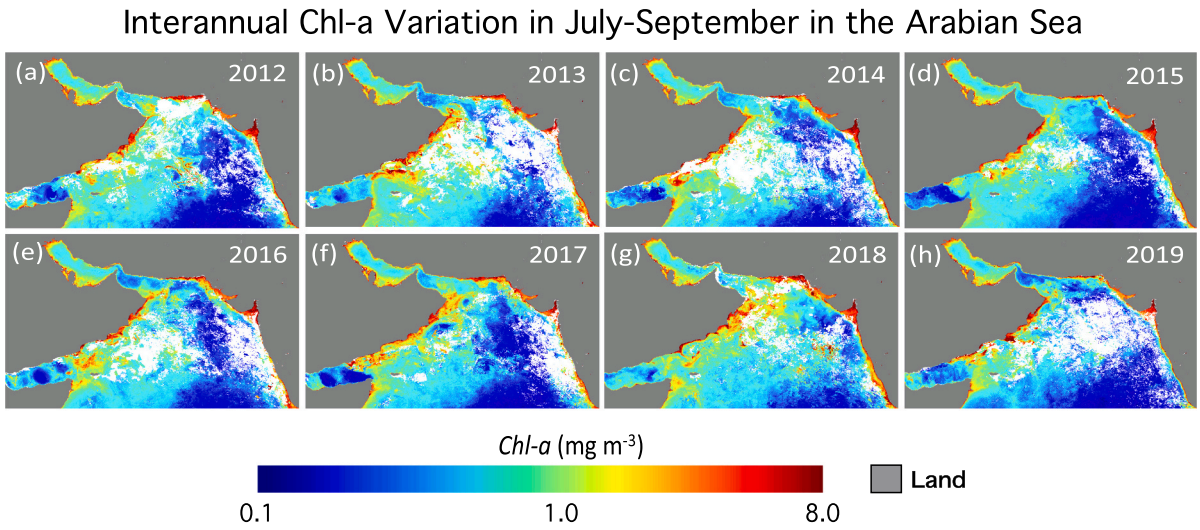


Fig. 6. Chl-a maps during the SW monsoon in July–September in the Arabian Sea for the year of (a) 2012, (b) 2013, (c) 2014, (d) 2015, (e) 2016, (f) 2017, (g) 2018, and (h) 2019.

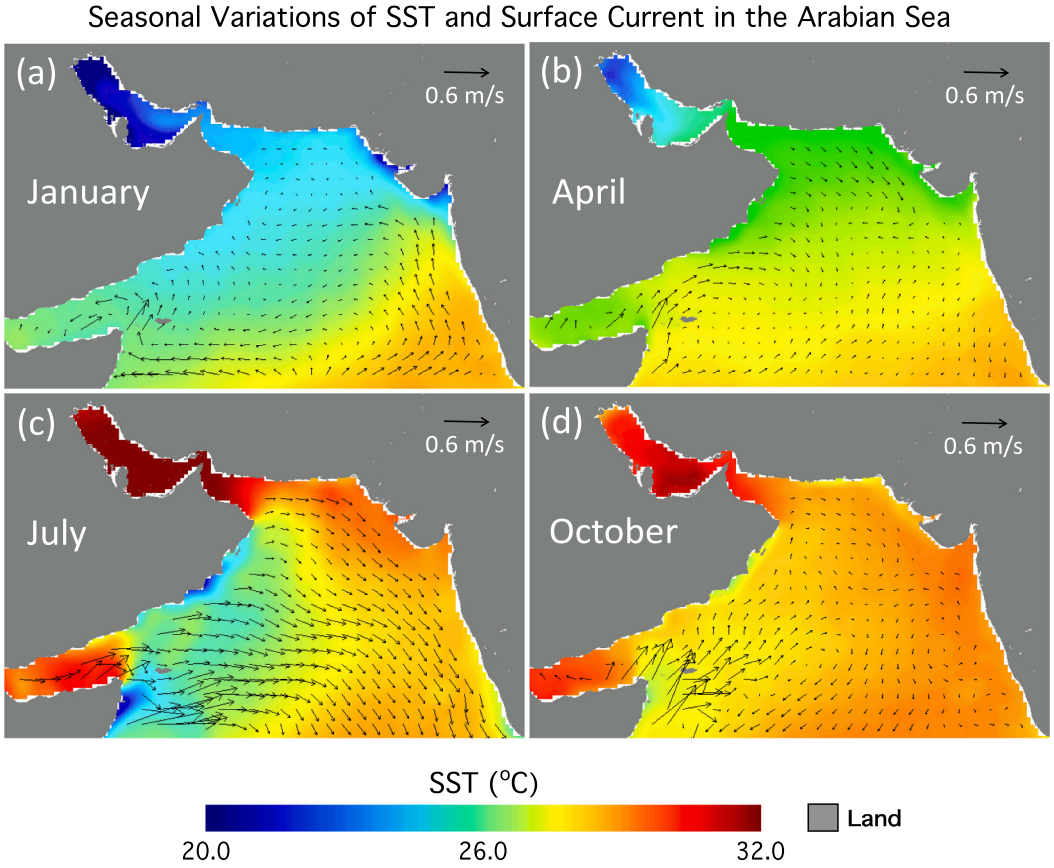


Fig. 7. Seasonal variation of the SST and surface current between 2012 and 2019 in the Arabian Sea for the month of (a) January (b) April, (c) July, and (d) October.

during the SW monsoon destroys the stratifications and leads to the moderately high MLD in these regions (Fig. 8k). On the other hand, the high MLD to the east of the Findlater Jet is driven by the downwelling and convergence of the surface water in this region. During the fall intermonsoon, the MLD drops to ~30–40 m (Fig. 8l) in the Arabian Sea due to the enhanced surface layer stratification following the diminished SW monsoon that drives upwelling and net heat gain in the upper layer of the Arabian Sea.

The interannual variabilities of $nL_w(443)$ (Fig. 9a) and Chl-a (Fig. 9b) in Box A of the western Arabian Sea and Box B of the northern Arabian Sea clearly show that the seasonal variabilities in these two regions are dominant in comparison to the interannual variability. Note that phytoplankton bloom is featured with decreased $nL_w(\lambda)$ at 443 nm. Thus, the elevated Chl-a comes with the depressed $nL_w(443)$ and vice versa. In these two regions, Chl-a ranges from the low value of ~0.15 mg m^{-3} to the high of over ~1 mg m^{-3}. On the other hand, the Chl-a seasonal variabilities for these two regions are not in phase with each other. Specifically, Chl-a in Box A peaks during the SW monsoon, while there

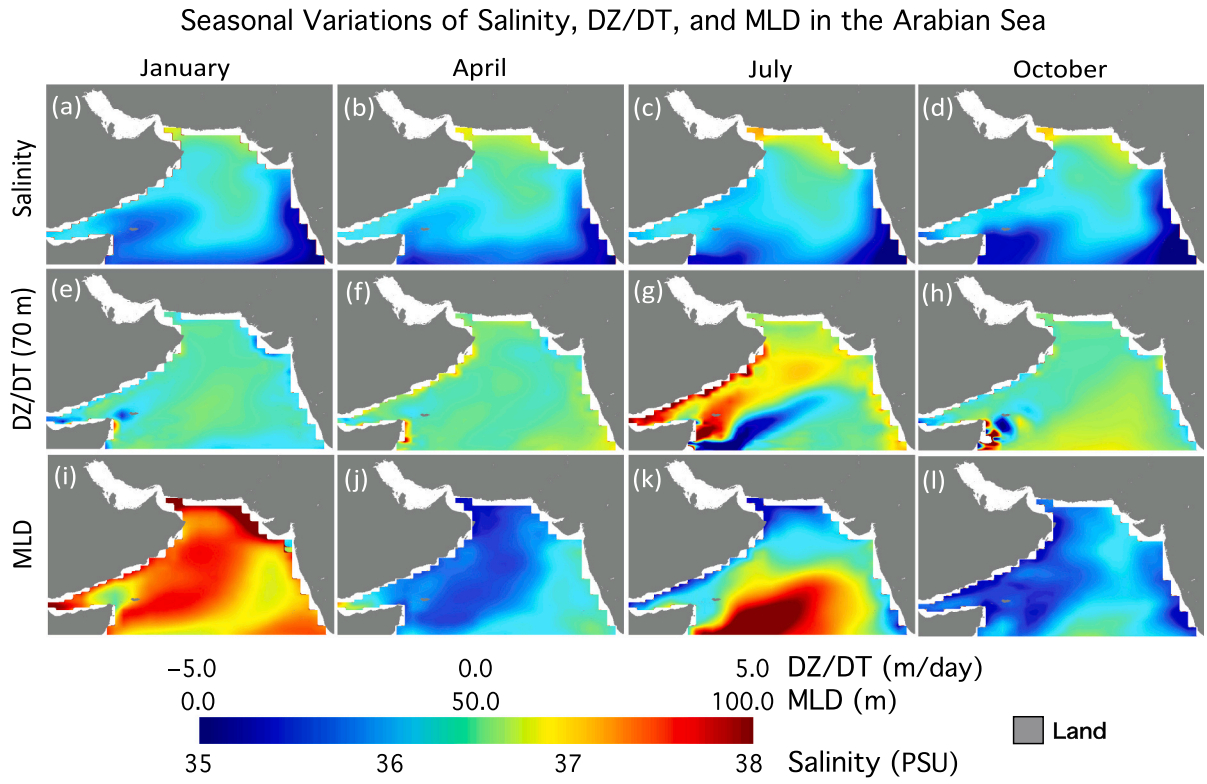


Fig. 8. Seasonal variation of SSS, vertical velocity (DZ/DT) at 70 m depth, and mixed-layer depth (MLD) in January, April, July, and October between 2012 and 2019 in the Arabian Sea for (a–d) SSS, (e–h) DZ/DT at 70 m, and (i–l) MLD.

are two Chl-a peaks in Box B with the major one in the NE monsoon and a small peak in the SW monsoon (Fig. 9b). The most significant change of Chl-a in Box A during the SW monsoon occurred in 2016. In that year, Chl-a in Box A jumped from <0.13 mg m^{-3} in May 2016 to 1.88 mg m^{-3} in August 2016.

The SST time series show two peaks and two troughs in a year for both regions (Fig. 9c). SST ranged between 24°C and 32°C between 2012 and 2020. The highest SST occurred in the spring intermonsoon with high SST in Box B lagging the high SST in Box A for about one month. In Box A, SST dropped from 30.77°C in May 2016 to 24.51°C in August 2016. The sharp SST drop occurred at the same time as a jump of Chl-a (Fig. 9b). In comparison to the significant SST drop in Box A, SST drop for Box B in the SW monsoon is generally less than half of that in Box A. During the NE monsoon, SSTs in both of Boxes A and B dropped to the lowest in a year. SST in Box B is generally ~0.5–1.0°C lower than that in Box A due to the stronger surface cooling and convective mixing in the northern Arabian Sea. On the other hand, SSS in Box A is generally 0.5 psu less than that in Box B (Fig. 9d). In comparison to the seasonal changes of SST, the seasonal changes of SSS in both Boxes A and B are not significant in these two regions even though SSS in Box A indeed shows that SSS changes with the seasonal monsoon in the western Arabian Sea.

The vertical velocities in these two regions show that they reach maximum during the SW monsoon (Fig. 9e). In Box A, the upwelling at the 70 m depth could reach 1 m/day in July during the SW monsoon, while it is much smaller in Box B. The downward vertical velocity of ~0.2–0.3 m/day in Box A suggests that the convergence occurred in this region with the reversal of wind from the SW monsoon to NE monsoon. It is also notable that the vertical velocity is only slightly negative in Box B during NE monsoon. This could be attributed to the convective mixing in the northern Arabian Sea. The MLDs in Box A and Box B have similar seasonal changes (Fig. 9f). The MLD in the northern Arabian Sea reaches over ~80 m in Box B in February. During the spring and the fall intermonsoons, the MLD in both Box A and Box B are ~20–30 m. In Box A, a mini-peak of ~40–50 m normally occurs in the SW monsoon due to the strong monsoon wind-driven upwelling in this region.

Using the monthly potential temperature, salinity, and vertical velocity profiles between 2012 and 2019, we compute the climatology potential temperature, salinity, and DT/DZ profiles in both Box A and Box B for various months. The profiles of these hydrographic parameters provide more details of the physical processes in both Box A and Box B (Fig. 10). In Box A, the seasonal change of the potential temperature reaches to more than 300 m depth (Fig. 10a). By comparison, in Box B the seasonal change of the potential temperature mostly occurs in the upper 100 m (Fig. 10b). This reflects the fact that the deep ocean water in Box A is brought to the sea surface due to the strong upwelling in this region during the SW monsoon, while only waters in the upper thermocline depth are mixed during the NE monsoon in Box B. In Box A, the profiles of the potential temperature in July show a notable difference from those in the spring and fall intermonsoons (Fig. 10a). However, the potential temperature profile in April is similar to that in October shown in Box B (Fig. 10b).

The salinity profiles in both regions are not as significant as the potential temperature profiles (Fig. 10c and d). The seasonal change of the salinity profile in Box A is more significant than that in Box B. It is also noted that the maximum salinity in Box A is located at the 70 m depth in January instead of the sea surface in the other seasons. The profiles of the vertical velocity show that the upwelling velocity in Box A in July peaks at ~1 m/day at the 70 m depth, and it gradually decreases to <0.1 m/s at the 10 m depth (Fig. 10e). Even at the depth of 500 m, the upwelling velocity is still >0.5 m/day. In Box B, the upwelling velocity is ~0.35 m/day at about 40–50 m depth, and increases to 0.5 m/day at 500 m depth (Fig. 10f). In Box A, the vertical velocity reaches maximum at −0.3 m/day at the 70 m depth, and it became −0.03 m/day at the 10 m depth in January. In Box B, the vertical velocities in January, April, and October are insignificant and close to 0 for the entire water column.

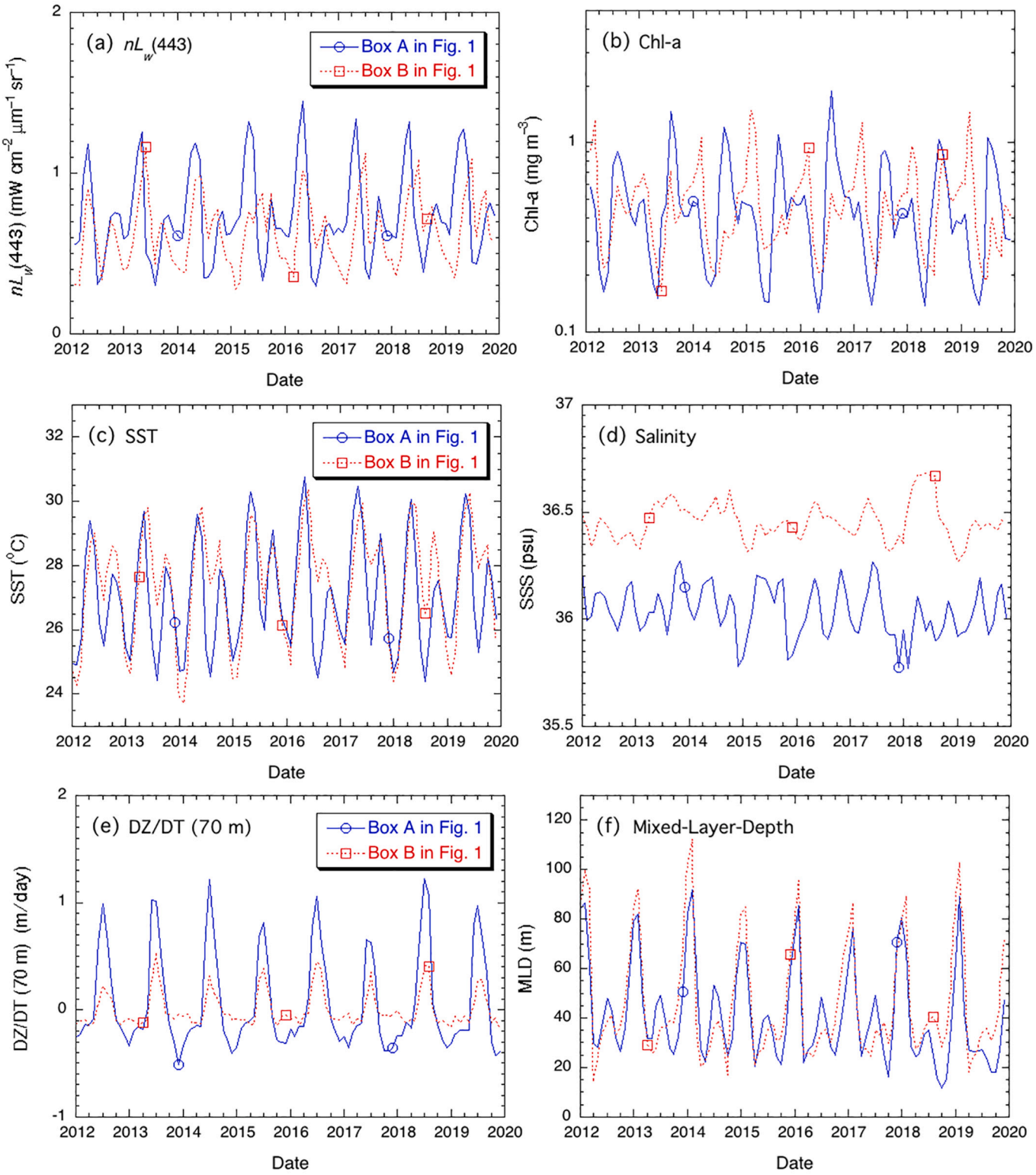


Fig. 9. Temporal variations in Box A and Box B between 2012 and 2019 in the Arabian Sea for (a) nL_w(443), (b) Chl-a, (c) SST, (d) SSS, (e) DZ/DT at 70 m, and (f) MLD.

3.3. Seasonal nutrient dynamics in the Arabian Sea

The surface distributions of nutrients such as nitrate (Fig. 11) and phosphate (Fig. 12) are highly dynamic and driven by seasonal monsoon winds in the Arabian Sea. In January, enhanced nitrate can be observed in the northern Arabian Sea (Fig. 11a). In February and March, the nutrient levels in the northern Arabian Sea are elevated (Fig. 11b and c). In the intermonsoon in April (Fig. 11d) and May (Fig. 11e), low nitrate concentrations cover the entire Arabian Sea. In June, only a slight increase of the nitrate concentration can be observed along the western Arabian Sea coast with the onset of the SW monsoon (Fig. 11f). In July (Fig. 11g), August (Fig. 11h), and September (Fig. 11i), the Arabian Sea features the highest nitrate concentration over 10 μmol l^{-1} along the Somalia and Arabian Peninsula coastal regions. In August and

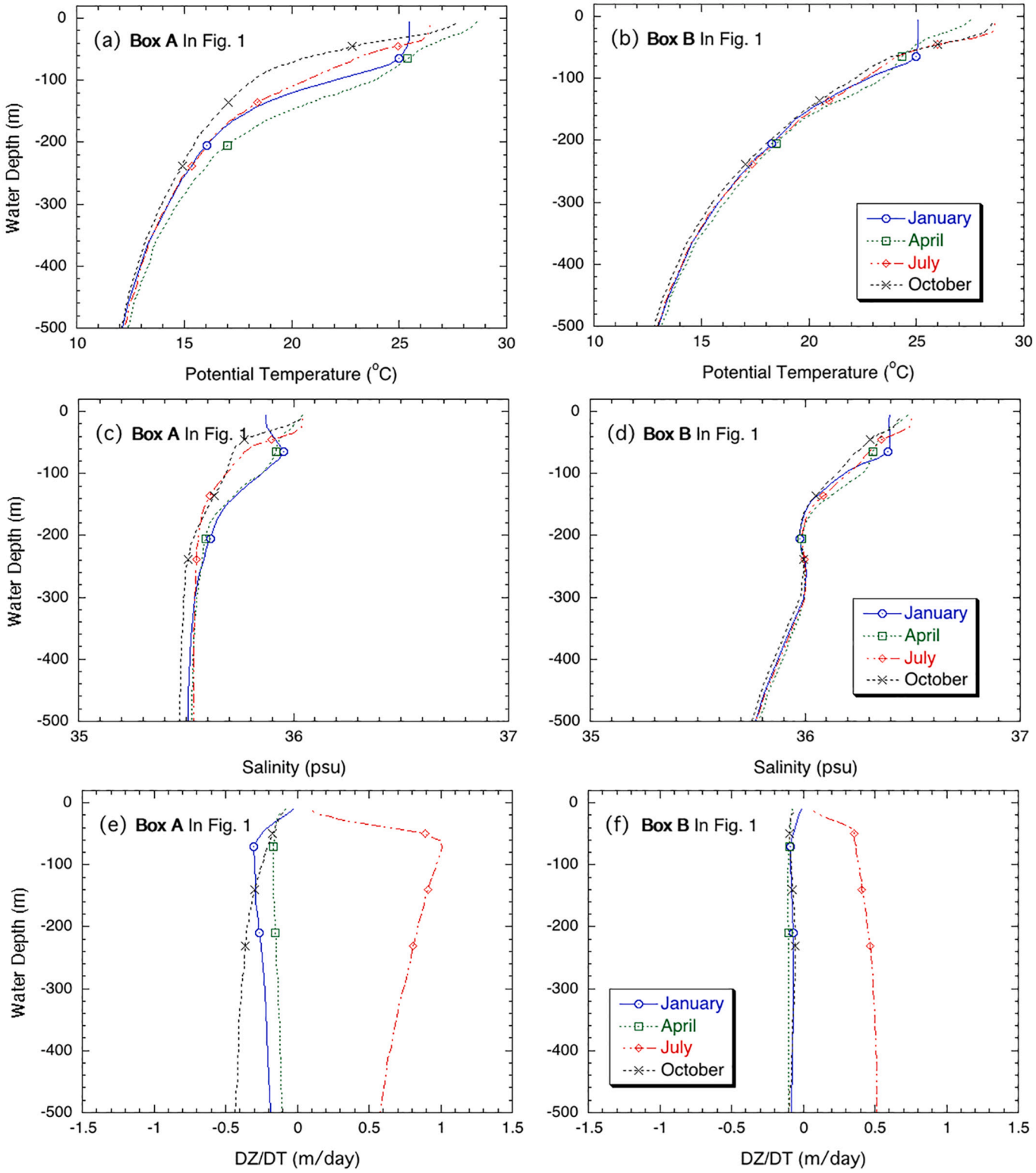


Fig. 10. Seasonal variations of the profiles at the upper 500 m in Box A and Box B between 2012 and 2019 in the Arabian Sea for (a) potential temperature in Box A, (b) potential temperature in Box B, (c) salinity in Box A, (d) salinity in Box B, (e) DZ/DT in Box A, and (f) DZ/DT in Box B.

September, the area of high nitrate concentration can be observed to further extend offshore to the central Arabian Sea. In October, the nitrate concentration in the Arabian Sea drops significantly in the region (Fig. 11j), and it drops further in November (Fig. 11k). In fact, the low nitrate distribution in November is similar to that in May. In December (Fig. 11l), the nitrate concentration in the northern Arabian Sea begins to gradually increase.

Even though the seasonal variation of the phosphate concentration (Fig. 12) is generally similar to that of the nitrate concentration (Fig. 11) in the Arabian Sea, i.e., elevated nitrate and phosphate concentrations in the northern Arabian Sea during the NE monsoon with their highest values in the western Arabian Sea during the SW monsoon, there are

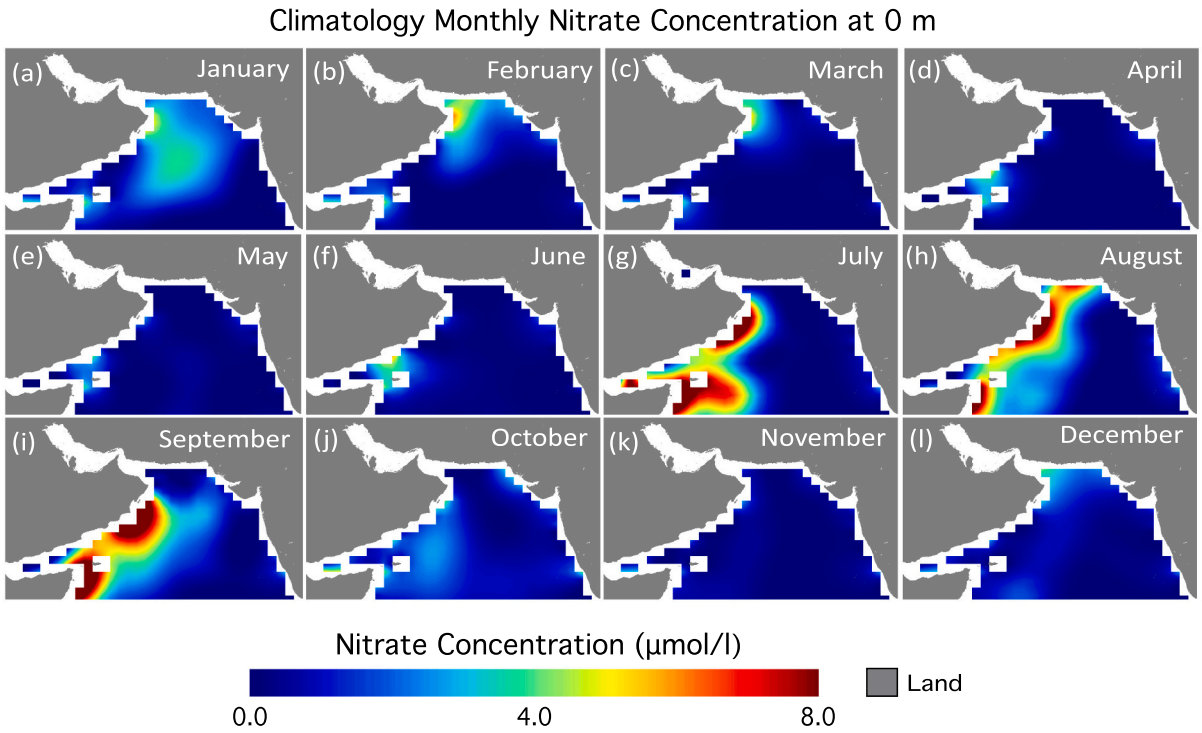


Fig. 11. Sea surface nitrate concentration in the Arabian Sea for the month of (a) January, (b) February, (c) March, (d) April, (e) May, (f) June, (g) July, (h) August, (i) September, (j) October, (k) November, and (l) December.

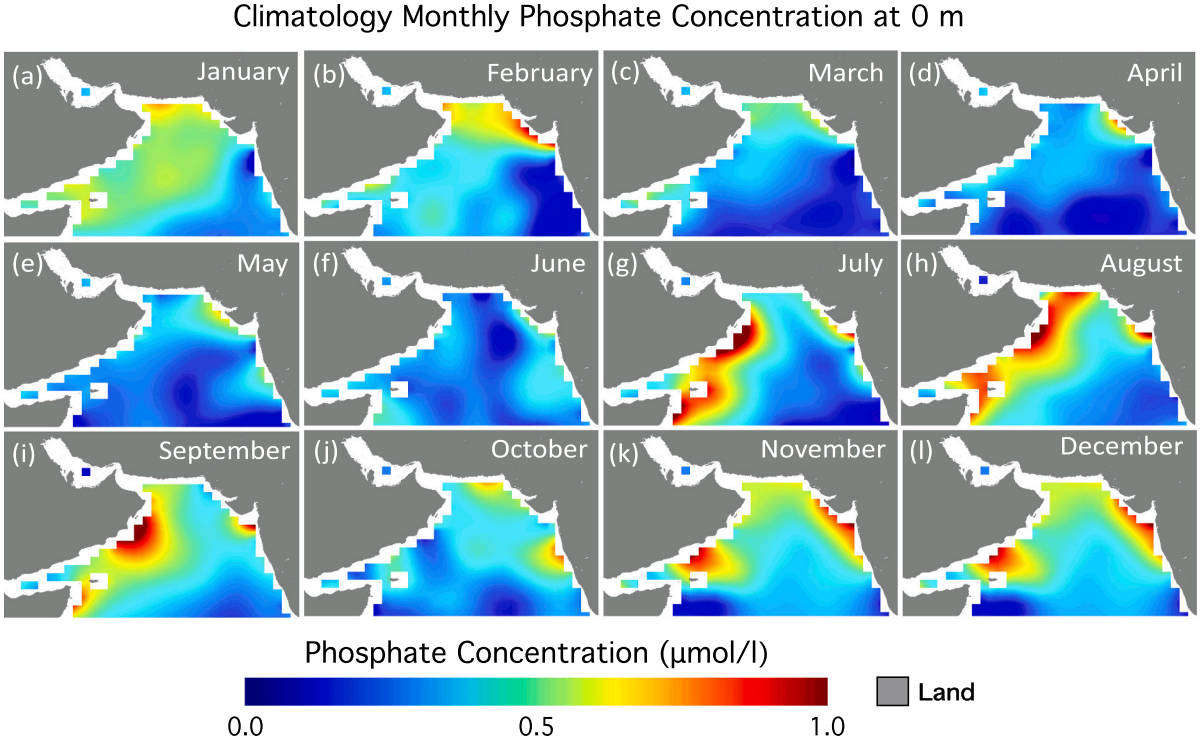


Fig. 12. Sea surface phosphate concentration in the Arabian Sea for the month of (a) January, (b) February, (c) March, (d) April, (e) May, (f) June, (g) July, (h) August, (i) September, (j) October, (k) November, and (l) December.

some differences in the seasonal dynamics of the nitrate and phosphate concentrations. In fact, the phosphate concentration in the Arabian Sea normally is only a fraction of the corresponding nitrate concentration. In January (Fig. 12a) and February (Fig. 12b), the phosphate concentration is elevated more broadly than the nitrate concentration in the Arabian Sea in the same months (Fig. 11a and b). In November and December, the phosphate concentration in most of the Arabian Sea is still moderately elevated (Fig. 12k and l). In comparison, however, the nitrate concentration is notably low in these two months (Fig. 11k and l).

The vertical profiles of nitrate and phosphate further show the

seasonal changes and the difference of nutrient vertical distributions in Boxes A and B (Fig. 13). In general, the depth with the sharp nitrate and phosphate concentration changes is at the upper 200 m in Box A (Fig. 13a and c). In Box B, however, the depth with sharp nitrate and phosphate concentration changes is at the upper ~100 m (Fig. 13b and d). Seasonal changes of the nitrate and phosphate concentrations in the upper layers are also significant. In the MLD of Box A, the nitrate concentration ranges from about 16 μmol l^{-1} at the depth of 50 m to about 4.7 μmol l^{-1} at the sea surface in July, while the nitrate concentration ranges from ~6.8 μmol l^{-1} at the depth of 80 m to ~0.47 μmol l^{-1} at the sea surface in January (Fig. 13a). The vertical profile of the phosphate concentration in the MLD of Box A ranges from ~1.13 μmol l^{-1} to ~0.42 μmol l^{-1} in July and from ~0.93 μmol l^{-1} to ~0.52 μmol l^{-1} in January (Fig. 13c). In Box B, the nitrate concentration is nearly uniform at ~3 μmol l^{-1} in January at the upper 60 m depth (Fig. 13b). This is significantly higher than nutrient concentrations in the other seasons. In April, the nitrate concentration at the upper 20 m depth is extremely low at <0.1 μmol l^{-1} in Box B. In comparison, phosphate concentrations in January are only slightly larger than those in the other seasons. Specifically, the phosphate concentration in January ranges from ~1.02 μmol l^{-1} at the 80 m depth to ~0.54 μmol l^{-1} at the surface, while the phosphate concentration ranges between ~0.38 μmol l^{-1} and ~ 1.16 μmol l^{-1} at the same depth range (Fig. 13d).

3.4. Chl-a climatology in the SeaWiFS era

It has been debatable on whether the trend of Chl-a in the Arabian Sea is increasing or decreasing. One study shows that warming of the Eurasian landmass has been making the Arabian Sea more productive. The increase was attributed to the escalation of the intensity of summer monsoon winds (Goes et al., 2005). However, another study shows that Chl-a concentration appears to be declining since 2003, and the declining Chl-a may be caused by the increasing sea level anomaly (SLA) in the region (Prakash et al., 2012).

The same (or similar) algorithms (or methodology) for satellite ocean color data processing (e.g., sensor vicarious calibration, atmospheric correction, retrievals of $nL_w(\lambda)$ spectra and Chl-a, etc.) from SeaWiFS and VIIRS determine that Chl-a products from these sensors have been continuous and consistent since 1997 (Franz et al., 2005; Wang et al., 2020). Thus, Chl-a data from SeaWiFS and VIIRS can be used to study the long-term changes of phytoplankton biomass in the Arabian Sea. Fig. 14 shows the SeaWiFS-derived Chl-a climatology from 1997 to 2010. Since both Chl-a climatology data from SeaWiFS and VIIRS are computed from multiple-year observations, the interannual variability of Chl-a is largely filtered out and its impact is minimized in each dataset when computing the climatology for these two sensors. Consequently, comparison of the two Chl-a climatology in these two periods indeed can provide some insights of two decades of Chl-a fields in the Arabian Sea.

Overall, the seasonal variability, spatial patterns and magnitudes of SeaWiFS Chl-a in each month shown in Fig. 14 are similar to the Chl-a

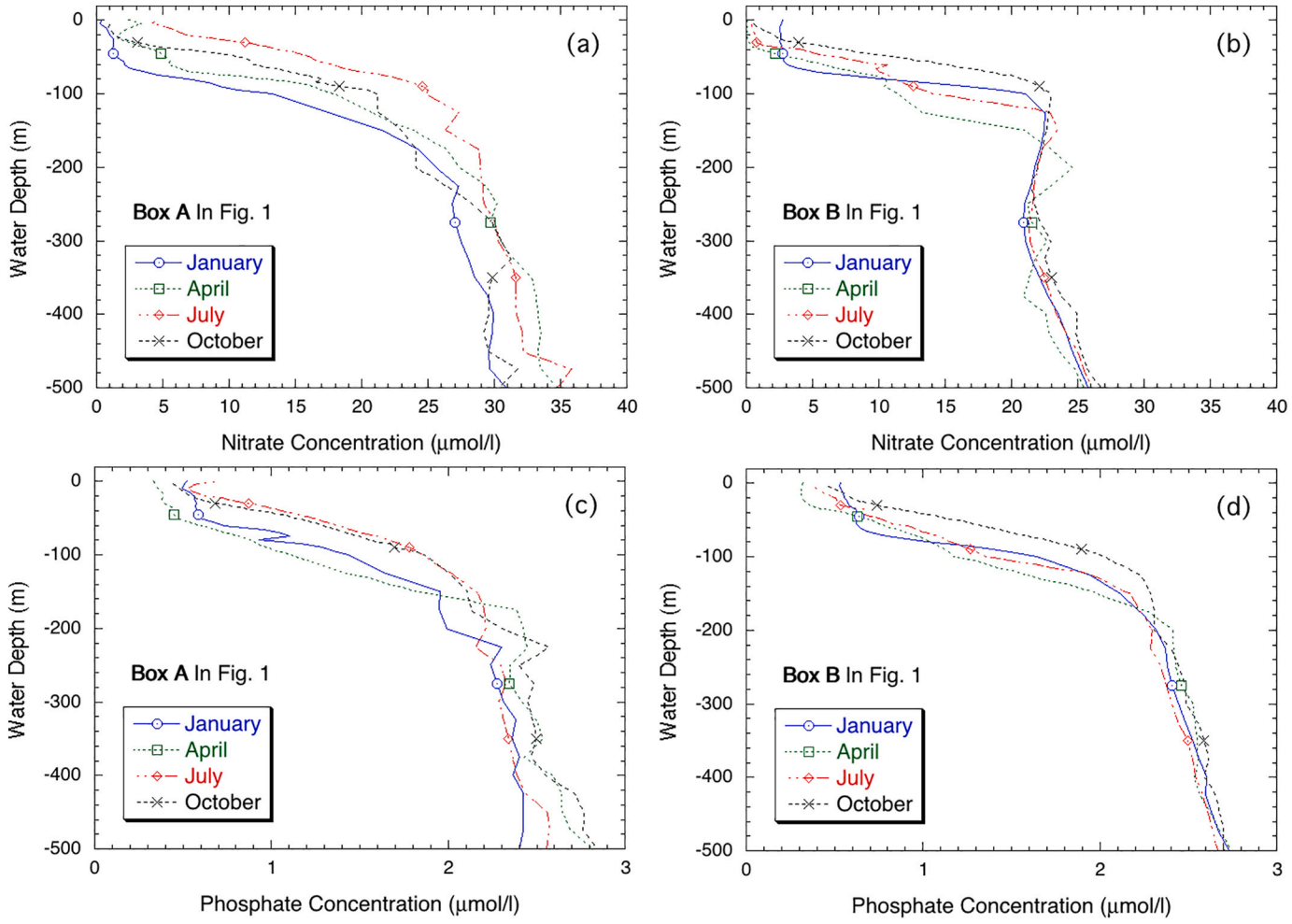


Fig. 13. Seasonal variations of the profiles at the upper 500 m in Box A and Box B between 2012 and 2019 in the Arabian Sea for (a) nitrate concentration in Box A, (b) nitrate concentration in Box B, (c) phosphate concentration in Box A, and (d) phosphate concentration in Box B.

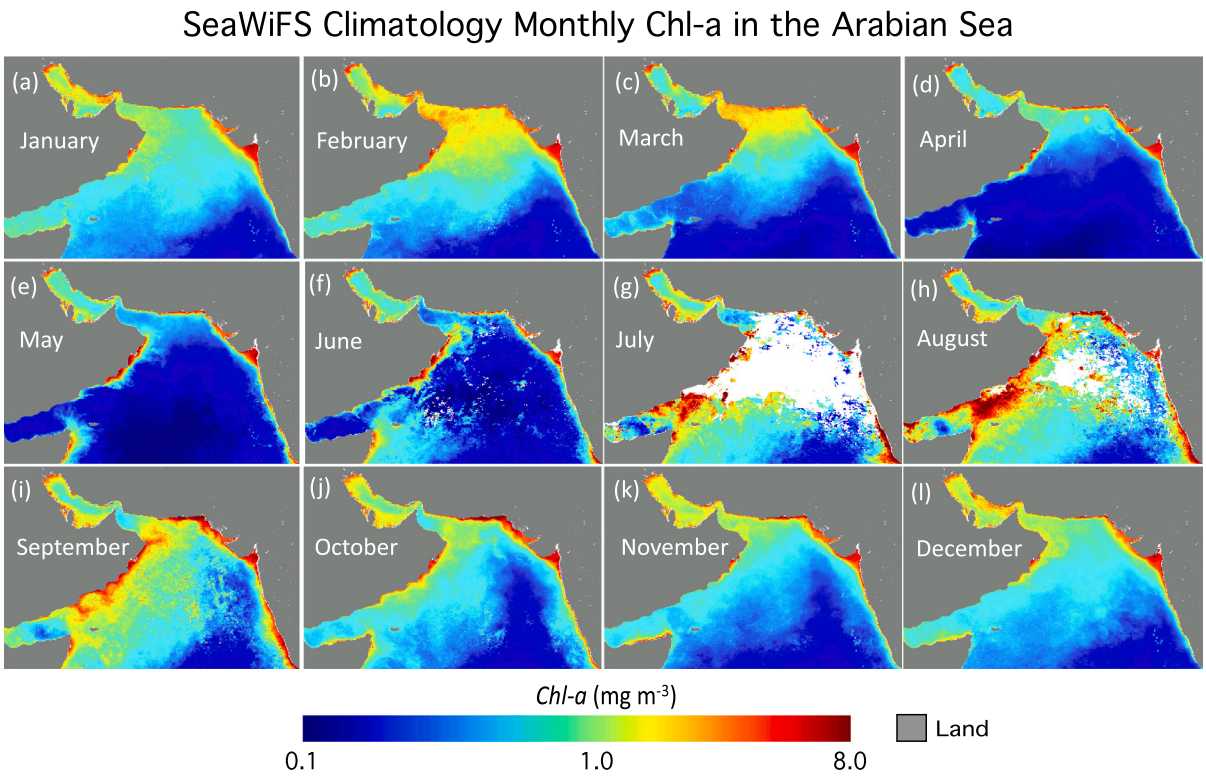


Fig. 14. Climatology monthly Chl-a between 1997 and 2010 in the Arabian Sea from the SeaWiFS observations for the month of (a) January, (b) February, (c) March, (d) April, (e) May, (f) June, (g) July, (h) August, (i) September, (j) October, (k) November, and (l) December.

climatology from VIIRS observations in Fig. 3. In each month, the mean ratio of the climatology Chl-a from VIIRS and SeaWiFS for the entire Arabian Sea is within about 1.0 ± 5%. This shows that the physical and biological processes driven by the seasonal monsoon are still the dominant forces that drive the phytoplankton biomass dynamics in the region. However, there are indeed some differences in monthly Chl-a dynamics between SeaWiFS and VIIRS. Even though both show that enhanced Chl-a were located in the northern Arabian Sea in the NE monsoon (Figs. 3a–c and 14a–c) with similar magnitudes of Chl-a, the coverage of the enhanced Chl-a from VIIRS observations (Fig. 3a–c) can be seen extending further south to the central Arabian Sea in comparison to the coverage of the enhanced Chl-a from SeaWiFS (Fig. 14a–c). Similarly, differences of the monthly climatology Chl-a between SeaWiFS and VIIRS also occur during the SW monsoon. These Chl-a differences can be attributed to the non-uniform changes of SST and SLA in these two periods as shown in Prakash et al. (2012). During the spring intermonsoon, however, the Arabian Sea became oligotrophic (i.e., low Chl-a) from both of these datasets.

In the SW monsoon, Chl-a in the upwelling zone along the Arabian Sea coast were more pronounced in the SeaWiFS era (Fig. 14g–i) than those in the VIIRS era (Fig. 3g–i). In addition, the enhanced Chl-a in the SW monsoon were more broadly distributed in the Arabian Sea from SeaWiFS observations than that from VIIRS measurements. This implies that the productivity in the Arabian Sea during the SW monsoon decreased in the last two decades. Therefore, the Chl-a results during the SW monsoon in the Arabian Sea from this work agree with the study from Prakash et al. (2012).

4. Discussion

As an indicator for the phytoplankton biomass, Chl-a retrievals from satellite observations in the Arabian Sea in this study are computed using the OCI or blue-green reflectance ratio Chl-a algorithms (Hu et al., 2012; O'Reilly et al., 1998; Wang and Son, 2016). These algorithms were tuned to have good accuracy for the global ocean. It generally works well in the Arabian Sea (Tilstone et al., 2011). This suggests that the results characterizing and quantifying the phytoplankton blooms in the Arabian Sea from this study are reliable.

The biannual phytoplankton blooms are a unique biological feature of the Arabian Sea. This is different from the phytoplankton blooms normally occurring in spring or autumn in most of the world ocean. Since the phytoplankton blooms are driven by the reversals of the monsoon wind in the Arabian Sea, it is essential to use multiple datasets to characterize and quantify the phytoplankton bloom and understand the physical and nutrient dynamics that drive the seasonal variability and the phytoplankton blooms during the SW and NE monsoons. In the Arabian Sea, most of the knowledge about the monsoon-driven physical, biological, and biogeochemical processes was acquired from various studies, particularly the US Joint Global Ocean Flux Study (JGOFS) Arabian Sea Process Study (ASPS) (Smith, 2001; Smith et al., 1998). In comparison to the previous studies, which were mostly based on the in situ measurements from the US JGOFS ASPS in the 1990s, this study uses comprehensive datasets including the satellite Chl-a data, monsoon wind data, the corresponding hydrographic data, i.e., SST, SSS, temperature and salinity profile, MLD, vertical velocity, as well as nutrient seasonal dynamics data to provide a thorough comprehension of the two phytoplankton blooms occurring annually in the Arabian Sea between 2012 and 2019, and the driving forcing of the phytoplankton blooms. Instead of providing the general mechanisms for the phytoplankton blooms in the region qualitatively, this study provides not only detailed evaluations of the season phytoplankton blooms, i.e., magnitude, spatial extent, seasonal and interannual variability, but also the quantitative assessment of the physical and nutrient forcing that drive the phytoplankton biomass dynamics in the Arabian Sea. In comparison to the previous studies, this study provides new/improved knowledge and understanding of the seasonal and interannual phytoplankton biomass dynamics, the corresponding physical processes, and nutrient dynamics in the Arabian Sea.

4.1. Phytoplankton biomass dynamics

In the US JGOFS ASPS period of 1990s, the Coastal Zone Color Scanner (CZCS), which was only functional between 1978 and 1986 after was the only available satellite ocean color sensor to provide the large-scale observations of phytoplankton pigment in the Arabian Sea. The seasonality of the phytoplankton pigment was indeed identified (Banse and English, 2000). However, the phytoplankton pigment retrievals from the CZCS were rudimentary and less accurate in comparison to the data from later satellite ocean color sensors due to various limitations, e.g., the instrument design and spectral band specification, sensor calibration, development of algorithms for atmospheric correction, ocean biological/biogeochemical products, etc. In fact, from CZCS observations, it was not clear how long the high phytoplankton productivity lasted during the SW monsoon (Banse and English, 2000).

Even though the Arabian Sea was indeed studied using satellite ocean color for the Chl-a variability with MODIS observations (Jayaram et al., 2018), this study is the only comprehensive study to use multiple datasets not only to quantify the phytoplankton biomass dynamics, but also characterize and assess the physical processes and nutrient dynamics that eventually drive the Chl-a variability in the region. Multi-year observations of Chl-a as shown in Figs. 3–6 provide the quantitative assessment of the seasonal and interannual variabilities in the Arabian Sea. They also provide a detailed assessment of the extent and coverage of the two phytoplankton blooms. In addition to Chl-a that represent the biological activity in the region, this study also shows the significant seasonal variability of $nL_w(\lambda)$ spectra with the reversal of the seasonal monsoon. This implies that other optical, biological, and biogeochemical properties, such as the water diffuse attenuation coefficient at 490 nm $K_d(490)$, and color dissolved organic matter (CDOM), ocean inherent optical properties (IOPs) (Lee et al., 2002; Shi and Wang, 2019), also experience a similar seasonal variability in the Arabian Sea due to the seasonal monsoon.

The comparison of the monthly Chl-a climatology from SeaWiFS (Fig. 14) and that from VIIRS (Fig. 3), as well as statistics in ratios of the VIIRS and SeaWiFS climatology Chl-a over the entire Arabian Sea from Table 1, show that overall no long-term trend of a Chl-a change in the entire Arabian Sea can be concluded in the last two decades even though there are some differences between these two climatology datasets. Chl-a ratio values in the Arabian Sea from SeaWiFS and VIIRS are mostly within about 5%. Specifically, Chl-a ratios are within the range of ~1–2% during the SW monsoon (Table 1). In particular, Chl-a ratio results in Tables 2 and 3 show some spatial differences in the Arabian Sea although Chl-a ratios between VIIRS and SeaWiFS in the western Arabian Sea (Box A, Table 2) and the northern Arabian Sea (Box B, Table 3) are generally similar to those from the entire region shown in Table 1. In fact, in the region of Box A, Chl-a were less pronounced slightly from VIIRS than those of two decades ago from SeaWiFS for the period between June and November (Table 2). On the other hand, slightly enhanced Chl-a in both the western and northern Arabian Sea regions during the NE monsoon were found from VIIRS observations (Tables 2 and 3). During the NE monsoon, Chl-a reached its maximum in March in the VIIRS period, while Chl-a in a large portion of the northern Arabian Sea reached maximum in February in the SeaWiFS climatology Chl-a data. However, these differences are insignificant in comparison to the climatology Chl-a values, and they may be attributed to the differences in the interannual variability of local winds and ocean dynamics in these two datasets.

In addition, the no long-term trend of Chl-a in the Arabian Sea also suggests that the primary productivity in the region during the VIIRS era is similar to that in the SeaWiFS era two decades ago. The wind fields during the VIIRS era as shown in Fig. 2 are also similar to those in the SeaWiFS period (not shown here). This further suggests that the seasonal monsoon driving forces for the Chl-a dynamics, nutrient dynamics, and ocean dynamics, in terms of both the spatial distribution and their magnitudes, are similar in the last 20 years.

On the other hand, even though a lag of 1–2 months between Chl-a blooms and the seasonal monsoon were reported (Bartolacci and Luther, 1999) in the US JGOFS ASPS, it was never quantified and characterized in detail. Here, the combined datasets of the monsoon winds, SST, MLD, etc., provide a clear linkage between the atmosphere and ocean processes and the phytoplankton blooms during the SW and NE monsoons. The NE monsoon starts in November and reaches peaks in December and January with the strongest along the Arabian coast. In comparison, the phytoplankton blooms in the northern Arabian Sea occurs in February and March (Fig. 3b and c). During the SW monsoon, the winds start in May, and the strong winds lasted from June to August. There is apparently a one-month lag of the phytoplankton bloom in the western and central Arabian Sea (Fig. 3f–h). In the specific regions shown in Box A and Box B in Fig. 1, detailed information is also provided about how the Chl-a variability is correlated with the changes of SST, salinity, DZ/DT, and MLD between 2012 and 2020 (Fig. 9), i.e., enhanced monthly Chl-a comes with lower SST and vice versa, etc. This

Table 1
Statistics of climatology $Chl\text{-}a^{VIIRS}/Chl\text{-}a^{SeaWiFS}$ (mean, median, and standard deviation (STD)) in the Arabian Sea.

Month	Mean Ratio	Median Ratio	STD
January	1.050	1.065	0.099
February	1.023	1.021	0.111
March	1.068	1.091	0.100
April	1.068	1.096	0.103
May	1.067	1.088	0.097
June	1.008	1.012	0.111
July	1.011	1.015	0.110
August	0.982	0.976	0.108
September	0.977	0.968	0.110
October	0.997	0.999	0.109
November	1.032	1.041	0.104
December	1.040	1.056	0.101

Table 2
Statistics of climatology $Chl\text{-}a^{VIIRS}/Chl\text{-}a^{SeaWiFS}$ (mean, median, and standard deviation (STD)) in Box A of Fig. 1.

Month	Mean Ratio	Median Ratio	STD
January	1.048	1.055	0.084
February	1.069	1.094	0.103
March	1.070	1.095	0.104
April	1.110	1.150	0.095
May	1.011	1.015	0.098
June	0.981	0.972	0.114
July	0.960	0.945	0.110
August	0.957	0.934	0.111
September	0.928	0.900	0.100
October	0.954	0.941	0.099
November	0.978	0.970	0.098
December	1.050	1.059	0.080

Table 3
Statistics of climatology $Chl\text{-}a^{VIIRS}/Chl\text{-}a^{SeaWiFS}$ (mean, median, and standard deviation (STD)) in Box B of Fig. 1.

Month	Mean Ratio	Median Ratio	STD
January	1.045	1.051	0.086
February	0.911	0.893	0.085
March	1.049	1.064	0.102
April	1.081	1.096	0.088
May	1.074	1.089	0.086
June	1.030	1.043	0.109
July	1.086	1.087	0.107
August	0.965	0.946	0.113
September	0.943	0.924	0.105
October	0.979	0.970	0.108
November	1.044	1.059	0.097
December	1.078	1.087	0.072

agrees with the other study (Prakash et al., 2012).

4.2. Physical, biological, and biogeochemical processes

The physical, biological, and biogeochemical processes that drive the seasonal variability of Chl-a in the Arabian Sea are consistent with the known mechanisms for the phytoplankton blooms during the SW and NE monsoons. The strong upwelling in the western Arabian Sea brings the cold high-nutrient deep ocean water into surface, and leads to a broad decrease of SST and increase of nitrate and phosphate in the ocean surface in the western Arabian Sea during the SW monsoon (Figs. 11g–i and 12g–i). The phytoplankton bloom during the NE monsoon is not as strong as that in summer (Fig. 6). It can be attributed to less powerful physical forcing and weak enhancement of the nutrient fields during the NE monsoon. All of these are supported by various studies in the US JGOFS ASPS (Bartolacci and Luther, 1999; Shi et al., 1999; Wiggert et al., 2005; Wiggert et al., 2000).

This study indeed provides the improved knowledge for the processes such as the upwelling, MLD, nutrient distributions and profiles in the Arabian Sea to better understand the physical, biological, and biogeochemical processes in the region. In the US JGOFS ASPS, the ocean physical, biological, and biogeochemical processes were analyzed and synthesized through the in situ measurements of the parameters such as the temperature, MLD, and nutrients for the upper ocean response to the monsoon (Dickey et al., 1998; Kumar and Narvekar, 2005; Morrison et al., 1998; Weller et al., 2002). With these scattered measurements from the moorings and field cruise experiments, it is impossible to further quantify the processes for different regions in the Arabian Sea. In contrast, this study provides both the spatial and vertical quantifications of these physical, biological, and biogeochemical parameters, and shows how these processes lead to the seasonal variability of Chl-a in the Arabian Sea. As an example, the maximum upwelling speed reached over 1 m/day at the depth range of 60–100 m (Fig. 10e). The upwelling velocity is still over 0.5 m/day even at 500 m depth.

Further evidence that the phytoplankton bloom occurs during the SW monsoon, which is driven by the coastal upwelling, offshore Ekman pumping, and the advection of the coastal waters offshore, is provided when the spatial pattern of SST drops (Fig. 7c) matches with the enhanced Chl-a pattern during the SW monsoon (Fig. 3g–i). During the NE monsoon, the vertical mixing due to the surface cooling can only penetrate to less than ~100 m as suggested by the MLD (Fig. 9f) and the potential temperature profiles (Fig. 10b) in Box B. This provides solid evidence why the phytoplankton bloom during the NE monsoon is not as strong as that in summer (Fig. 6) as reported in the US JGOFS ASPS (Kinkade et al., 2001).

4.3. Nutrient dynamics

In the Arabian Sea, the two distinct periods of elevated biological activity come with the surface layer nutrient enrichment that supports the increased rates of primary productivity (Wiggert et al., 2005). Variations in particulate organic carbon (POC) and Chl-a were greatly affected by nutrient concentrations (Gundersen et al., 1998). These were drawn from the various in situ measurements. However, the spatial and temporal variability of the nutrient fields in the entire Arabian Sea are still missing and not well characterized and quantified. The spatial and temporal variability of the nutrient fields is critical for us to understand the seasonal dynamics of phytoplankton biomass in the Arabian Sea. They can help us to further understand the issues, such as why the phytoplankton bloom is stronger during the SW monsoon than that during NE monsoon, and why Chl-a are much higher during the fall intermonsoon than that during the spring intermonsoon.

In the western Arabian Sea, the nitrate and phosphate concentrations show significant increase at the upper 200 m (Fig. 13a and c). In contrast, the nitrate and phosphate concentrations in the northern Arabian Sea could only increase modestly at the mixed-layer as shown in Figs. 13b, d, 11a–c, and 12a–c. As an example, the average nitrate concentration in Box B is ~2.7 μmol l^{-1} at the upper 50 m in January in comparison to ~11 μmol l^{-1} at the upper 50 m in July in Box A. The difference of the nutrient concentrations for these two blooms is consistent with satellite observations which show that the phytoplankton bloom in the NE monsoon is weaker than that in the SW monsoon.

The results of the physical and the nutrient dynamics can not only reveal the mechanisms for the seasonal duo blooms in the Arabian Sea, but also provide insight about Chl-a during the two intermonsoons in the region. Between the two seasonal phytoplankton blooms, there are two intermonsoon periods, i.e., spring intermonsoon and fall intermonsoon. In these two intermonsoon periods, the physical conditions are similar, i.e., broad SST increase (Fig. 7), negligible upwelling and vertical mixing (Figs. 8 and 10), and close MLD (Fig. 8). However, Chl-a in the fall intermonsoon (Fig. 3j–k) are notably higher than those in the spring intermonsoon (Fig. 3e–f). This could be attributed to the difference of the nutrient levels in these two intermonsoons. The nitrate concentrations in the spring intermonsoon (Fig. 11d–e) are much smaller than those in the fall intermonsoon (Fig. 11j–k). As an example, the nitrate at the sea surface is nearly depleted with a concentration of only ~0.073 μmol l^{-1} in April in comparison to the sea surface nitrate concentration of ~0.28 μmol l^{-1} in October in Box B. Lack of the nitrate nutrient during the spring intermonsoon makes the Arabian Sea become oligotrophic. On the other hand, the phosphate concentration in the spring intermonsoon (Fig. 12d–e) is comparable to that in the fall intermonsoon (Fig. 12j–k). This suggests that the phosphate is not a major constraint for the low Chl-a in the spring intermonsoon. Thus, the nitrate concentration plays a controlling role in the biological variability in the Arabian Sea. This agrees with the in situ measurements of Chl-a and nutrient dynamics in the Arabian Sea (Morrison et al., 1998).

5. Conclusion

Phytoplankton biomass dynamics in the Arabian Sea is very different from the world ocean with two phytoplankton blooms driven by the semiannual reversal of the monsoon. Even though there were numerous studies about the physical and biological processes in the Arabian Sea, this study uses comprehensive datasets including satellite Chl-a data, monsoon wind data, corresponding hydrographic data, i.e., SST, SSS, temperature and salinity profile, MLD, vertical velocity, as well as the nutrient seasonal dynamics data to provide a thorough and comprehensive investigation of the biannual phytoplankton blooms and their driving forces. It provides not only detailed evaluations of the seasonal phytoplankton blooms, i.e., magnitude, spatial extent, seasonal and interannual variability, but also the quantitative assessment of the physical and nutrient forcing that drives the phytoplankton biomass dynamics in the Arabian Sea. Indeed, this study shows the new/improved knowledge and an understanding of phytoplankton biomass dynamics, physical, biological, and biogeochemical processes, and nutrient dynamics in the Arabian Sea.

In this study, we use VIIRS and SeaWiFS ocean color observations, various hydrographic data, and nutrient data to characterize and quantify the duo phytoplankton blooms in the Arabian Sea and assess their corresponding driving forces. The phytoplankton bloom during the SW monsoon is stronger than that during the NE monsoon in terms of both Chl-a values and the extent of the bloom. The phytoplankton bloom during the SW monsoon mainly covers the western and central Arabian Sea, while the phytoplankton bloom during the NE monsoon is located in the northern Arabian Sea. The interannual variability of Chl-a in the Arabian Sea is less significant, compared to the seasonal variability.

No long-term trend of the Chl-a change is identified in the Arabian Sea when comparing Chl-a climatology data between SeaWiFS and VIIRS. When assessing SeaWiFS observations, we found a further enhanced and enlarged Chl-a climatology in the northern Arabian Sea from VIIRS observations during the NE monsoon. On the other hand,

Chl-a in the western Arabian Sea were less pronounced in the VIIRS era than those in the SeaWiFS era during the SW monsoon. This study does not uncover a trend of increasing ocean productivity in the Arabian Sea during the SW monsoon.

The physical parameters and the nutrient data show that the mechanism for the phytoplankton bloom during the SW monsoon is different from that during the NE monsoon. During the SW monsoon, the strong monsoon winds cause intense coastal upwelling and offshore upward Ekman pumping. The cold high-nutrient water from the thermocline depth and depths below is brought to the sea surface in the western Arabian Sea and advects further offshore to the central Arabian Sea with the surface ocean current. The enhanced nutrient level during the SW monsoon in the western Arabian Sea triggers and maintains the summer phytoplankton bloom. The bloom lasts until the end of the SW monsoon.

During the NE monsoon, the phytoplankton bloom is mainly confined to the northern Arabian Sea. The vertical mixing driven by the surface cooling is the physical driving force for this bloom. The MLD during the NE monsoon reaches ~80 m depth. The deepening MLD caused by the vertical mixing boosts nitrate and phosphate levels moderately at the sea surface, and leads to the phytoplankton bloom in the northern Arabian Sea during the NE monsoon. Comparisons of the seasonal Chl-a change and seasonal changes of the nitrate and phosphate in the Arabian Sea also suggest that nitrate is the main nutrient limiting factor that controls the phytoplankton growth and the seasonal change of Ch-a in the region.

Declaration of Competing Interest

None.

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